

KAZI EHSANUL KARIM

Molecular Simulation | Nanoscale Transport | Multiscale Thermo-Fluids | Machine-Learning Interatomic Potentials

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Postdoctoral / Research Fellow Opportunities | Target start: early 2027

RESEARCH PROFILE

Computational researcher and Ph.D. candidate in Mechanical Engineering at the University of Ulsan, with first-author publications in Physical Review E and Physical Chemistry Chemical Physics. Combines molecular dynamics, statistical mechanics and continuum modelling to investigate mass and energy transport under nanoscale confinement. Current independent work builds and tests machine-learned interatomic potentials for CNT-confined NaCl–water, with a focus on diagnosing model error. Research interests include error-guided active learning, physics-informed modelling and agentic simulation workflows.

Research methods: LAMMPS EMD/NEMD; CP2K; DP-GEN / DeePMD-kit; Green–Kubo transport; continuum boundary conditions; Python / Linux / HPC; ANSYS thermo-fluids.

EDUCATION

University of Ulsan, Republic of Korea

Mar 2019 – Present

Ph.D. Candidate, Integrated M.S.–Ph.D. in Mechanical Engineering | Micro/Nano-scale Thermo-Fluids Engineering

Laboratory | Supervisor: Prof. BoHung Kim

Dissertation defense planned: November 2026 | Formal degree conferral expected: February 2027

Dissertation: What Survives Sub-nm Confinement? — The Meaning of Slip Length under Molecular Accessibility, Ergodicity, and Structure–Transport Decoupling.

Khulna University of Engineering & Technology (KUET), Bangladesh

May 2015

B.Sc., Mechanical Engineering

DOCTORAL RESEARCH EXPERIENCE

Doctoral Researcher — Molecular Simulation & Nanofluidics, University of Ulsan

Mar 2019 – Present

- **Boundary thermodynamics** — Developed a first-law energy-balance analysis of the nanoscale flow boundary and tested NEMD against Hagen–Poiseuille limits under controlled wall–fluid interaction (Physical Review E, 2021).
- **Water transport in CNTs** — Critically assessed scale effects and boundary definitions in CNT water-transport models, relating molecular-simulation evidence to continuum predictions (Physical Chemistry Chemical Physics, 2024).
- **Confinement and transport** — Investigate molecular accessibility, layering, finite-time diffusivity, hydraulic resistance and continuum-closure limits in carbon nanotubes and graphene nanoslits.
- **Reproducibility and uncertainty** — Apply ensemble, timestep, box-size, forcing and sampling checks; block averaging and confidence intervals; and thermostat-sensitivity and cross-method comparisons to test transport interpretations.

INDEPENDENT RESEARCH PROJECT

Error-Aware MLIPs for CNT-Confined NaCl–Water

2026 – Present

- **Workflow** — Built and ran an end-to-end workflow for NaCl–water in four carbon-nanotube diameters — CP2K (PBE-D3(BJ)) ab initio MD seeds and reference labels → DP-GEN → DeePMD-kit → LAMMPS — with a verification step at every stage.
- **Evaluation** — Developed species-resolved force validation on fresh-DFT holdout frames; identified a species-specific failure mode that the aggregate error concealed and committee disagreement did not flag; tested targeted reference-data augmentation (error unchanged; cause under investigation).
- **Transport** — Computed equilibrium (Green–Kubo) interfacial friction from MLIP-driven MD with block-bootstrap uncertainty and model-disagreement screening of trajectories; two standard estimators disagree, and the cause is under investigation.
- **Tooling** — Wrote a tested Python tool-kit for per-species force metrics, group-aware uncertainty, model-scope checks and rules-based decisions; public methods demonstration (synthetic data) at [kazi21.github.io](#).

PUBLICATIONS & MANUSCRIPTS

Peer-reviewed journal articles

1. K. E. Karim, M. Barisik, C. Bakli, B. Kim, “Estimating water transport in carbon nanotubes: A critical review and inclusion of scale effects,” Physical Chemistry Chemical Physics, 2024, 26(28), 19069–19082. [DOI](#)
2. K. E. Karim, B. Kim, “First law of thermodynamics on the boundary for flow through a carbon nanotube,” Physical Review E, 2021, 103(5), 053115. [DOI](#)

Under review / completed

3. K. E. Karim, B. Kim, "Molecular admission and packing govern transport in subnanometre graphene slits," under review.
4. K. E. Karim, B. Kim, "Wall-fluid interaction strength shifts the slit width of maximum finite-time diffusivity in a graphene-confined simple fluid," manuscript completed; preparing submission.
5. K. E. Karim, B. Kim, "Confinement-Dependent Hydraulic Resistance and Limits of Navier-Slip Transferability in Graphene Nanoslits," manuscript in final preparation.

In preparation — current research directions

Physics-informed neural networks (PINNs) for structure-friction-flow closure; multimodal electrolyte transport in carbon nanotubes; and error-aware machine-learning interatomic potentials (MLIPs) for agentic multiscale simulation.

Additional peer-reviewed work

A. S. M. M. Rahman, K. E. Karim, M. H. S. Simanto, "Effect of heat treatment on low carbon steel: An experimental investigation," *Applied Mechanics and Materials*, 2016, 860, 7–12. [DOI](#)

Also: one additional journal article on RPR manipulator modelling and five peer-reviewed conference papers in thermal, materials, and design engineering.

TECHNICAL EXPERTISE

- **Molecular simulation** — LAMMPS; EMD/NEMD; NVT/NPT/NVE; confined fluids; pressure-driven and Couette flow; Green-Kubo and NEMD transport coefficients; interfacial friction/slip; spatial binning and profile analysis.
- **Electronic structure & MLIP** — CP2K periodic GPW reference calculations using PBE-D3(BJ), GTH pseudopotentials and DZVP-MOLOPT basis sets; DP-GEN active learning; DeePMD-kit dataset curation, committee training, holdout validation, error analysis, stability testing and LAMMPS deployment.
- **Scientific computing & statistics** — Python (NumPy, SciPy, pandas, Matplotlib, MDAnalysis, scikit-learn), Bash, Linux/HPC, MPI workflows, automated parameter sweeps; block averaging, confidence intervals, convergence/sensitivity diagnostics, uncertainty-aware regression.
- **CFD & continuum modelling** — ANSYS geometry/mesh generation, compressible nozzle flow, boundary-condition specification, convergence monitoring, and pressure/density/temperature/velocity/Mach post-processing; nozzle-wall and coolant-passage thermal analysis.

ADDITIONAL PROJECTS & EXPERIENCE

Thermo-Fluid Simulation of a Methane Rocket-Engine Nozzle & Cooling Passage

2022

Industry-linked laboratory project using ANSYS for geometry/mesh generation, compressible nozzle-flow and high-temperature thermal analysis, and post-processing of density, pressure, temperature, velocity, and Mach-number fields; included nozzle-wall and methane coolant-passage thermal behaviour.

Research Assistant — Robotics & Mechatronics Laboratory, KUET

Feb 2014 – May 2015

Kinematic modelling of an RPR manipulator, automated-wheelchair design, and experimental thermal/materials studies.

TEACHING & MENTORING

Practical MLIP Crash Course (CP2K → DeePMD-kit → LAMMPS) — course developer

2026 – Present

- **Curriculum** — Developing a 30-lesson practical course for senior undergraduates and first-year M.S. students: runnable recipes, clearly labelled synthetic examples, a break-it-and-fix-it case in every lesson (worked solutions complete for six core lessons) and an interview playbook; learner pilot planned.

REFERENCES

References available upon request.